



# MISUS REFRACTORIES & ALLIED INDUSTRIES

ISO 9001:2015 & ISO 45001:2018 Certified

कर्मणां सिद्धिः धैर्येण लभ्यते।

Success in action is achieved through patience





**Monojit Sengupta (CEO)**

## From CEO's Desk

In The year 2004 this Refractory Industry (**Misus refractory & Allied Industries**) was founded by Late Manik Sengupta by his hands on touch with a high progressive visionaries. Misus Refractories & Allied Industries (MRAI) since 2004 steadily maintains its growth both in size and diversity to stand today in the radically changed economic and unprecedented competitive market of Refractory Monolithic.

The main focus was to fulfil special requirements of primarily steel industries and special applications. MRAI is committed to produce high-quality Mudgun mass through its state of art manufacturing facilities located at Barjora, Bankura, W.B. Growth is fundamental to human progress. It is about improving from present for a bright future. We should follow it as our guiding philosophy. After all, we want to live a better life and are in need of modern & improved technology. MRAI wants to strengthen its growth, so that our dream is fulfilled by adopting latest technology, Equipped with reliable machinery to produce high quality products that meet the needs of our customers.

That is why, we are trying to expand our technical capacity to deliver most advanced integrated system for manufacturing better quality Refractory Products suitable for latest modified Blast Furnaces of various Steel Plants in our country and fulfil customer's requirements and maintain Customer Satisfaction.

We have a team of experienced and skilled professionals, Who ensures that our products meet the stringent quality standards in state of art laboratory before delivering it to our clients.

We have excellent R&D facilities with all testing facilities for Refractory Materials. In fine may express that we dream of a company which not only produce good quality products but also mark its name as one of the best manufacturing companies of Refractory Products in India. We hope, we will reach the goal by adopting modern technology with the help of latest R&D facilities, skilled & experienced Technical Experts. There is no doubt that our company (**MRAI**) is participating part and partial to growth of our country's economy & progress.

### **Vision :**

Our Vision is to be a Leading, Competitive producer of top quality castable refractories & monolithic products and sound business practices to ensure customer satisfaction.

Our vision is to maintain corporate excellence, Technology upgradation, modern management system, Continual improvement of product quality and sustainable plant operation.

### **Mission :**

Our Mission is to strengthen the growth of **MRAI** supplying Quality Products to our valued customers and creates benchmark for quality on a global market through business excellence and Technology Upgradation.

### **Quality Policy :**

**Misus Refractories & Allied Industries** engaged in the manufacturing of Refractory Monolithic is dedicated to its commitment towards its quality and to make it a Hallmark of every process and activity for achieving satisfaction of all our Customers.

To maintain a quality management and assurance system of International standard and progressively create centres of excellence across organisation. Adopt continual improvement of process and products through innovation.

MRAI is a ISO 9001:2015 & ISO 45001:2018 Certified Company.

## ABOUT US :

The **Misus Refractories & Allied Industries** is one of the leading Refractory Monolithic Manufacturers in India equipped with all the modern facilities to manufacture high performance products. We provide a complete range of refractory solutions to the entire steel manufacturing unit.

The **Misus Refractories & Allied Industries** is a partnership organization committed to strengthen its growth by continuous Research & Development work & scattering its product quality & after service to all the valuable customers. Our Research & Development effort underscore our overall quality consciousness and aims at establishing technological leadership for our products & services in the market.

The **Misus Refractories & Allied Industries** is situated at Barjora, Bankura District in West Bengal on State Highway No. 9 and is located at a distance of 9 KM south of Durgapur Station in Eastern Railways. The factory is well connected by road and rail to all over India as well as nearest port at Kolkata and Haldia. With this connectivity our company can easily Transport/Deliver Refractory Products to steel plants & other plants all over India and abroad. It was established with the objective to produce high quality Monolithic Refractory like Anhydrous Mudgun Mass, Anhydrous Trough Mass, and Resin Bonded Trough Mass for Steel, Cement, Glass, Foundry, Petro Chemical, Aluminum Industries etc in India. The Misus Refractories & Allied Industries has maintained its leadership position in key sector of steel. From the beginning we have continuously invested in R & D and simultaneously modernized our production lines with focus on improving the quality of our product and better service.

### COMPANY'S MILESTONE :

| Year | Activities                                                                                                                                                                                                         |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2004 | Foundation.                                                                                                                                                                                                        |
| 2006 | Enhancement of Production Capacity by adding New Muller Mixer.                                                                                                                                                     |
| 2012 | Enhancement of Production Capacity by adding New Muller Mixer.                                                                                                                                                     |
| 2019 | Enhancement of Production Capacity adding Improved design Muller Mixer & Automatic Extruder machine.                                                                                                               |
| 2025 | Enhancement of Production Capacity adding Improved design Muller Mixer & Automatic upgraded Extruder machine (Ex-1 & Ex-2) & wrapping machine. Renovation of office building, plant infrastructure & weigh bridge. |

## New Project

M/s Misus Refractories & Allied Industries had undertaken a new expansion project of Mudgun/Taphole Mass at same site of capacity 1200MT / Month. With this expansion the Total plant capacity becomes 2400 MT / Month. All new machineries were installed with upgraded technology and with heating arrangement.

The Muller mixer and its conveyor belt was supplied by M/S Shreema Engineering Co., Kolkata and Extruder machine was supplied by M/S AMCL, Nagpur & Neptune Engg, Ahmedabad. The finished product from the extruder is wrapped using modern wrapping machine supplied by Sealed Air, Mumbai with high quality plastic wrapping material(Shrink film) which is 100 % recyclable and all CPCB norms are complied.

The raw material and finished goods storage area had been increased to store sufficient quantity of material. One additional vertical tar tank had been installed. Also renovation of office building & plant infrastructure was done.



## OUR PRODUCT

### Castable :-

The MR-Crete Range of Conventional Castables are generally usable for moderate and high heat Applications. These are generally Calcium Aluminate bonded Refractory for application in temperatures upto 1400°C High alumina conventional Castables are made with alumina aggregate and high alumina Refractory Cement. Suitable for application temperature to 1700° C. Low Cement Castables are available from 45% to 90%  $Al_2O_3$  content. This is suitable upto temperature 1800° C. It is proved because low porosity, better corrosion and erosion resistance, good thermal shock resistance. The wide range of insulating castables all available for both hot face and backup Refractory lining.

### Conventinal Castables

| Product         | Max service Temp°C | Water addition | Dry Density gm/cc | CC5(kgf/cm²) |        | Al <sub>2</sub> O <sub>3</sub> % | Fe <sub>2</sub> O <sub>3</sub> % |
|-----------------|--------------------|----------------|-------------------|--------------|--------|----------------------------------|----------------------------------|
| MRCRETE NORMAL  | 1400               | 11-12          | 2.10              | 110°C/24 hr  | 250.00 | 45.0                             | 4.0                              |
|                 |                    |                |                   | 900°C/2hr    | 200.00 |                                  |                                  |
|                 |                    |                |                   | 1300°C/2hr   | 270.00 |                                  |                                  |
| MRCRETE SUPER   | 1450               | 11-12          | 2.45              | 110°C/24hr   | 350.00 | 66.0                             | 5.0                              |
|                 |                    |                |                   | 900°C/2hr    | 300.00 |                                  |                                  |
|                 |                    |                |                   | 1400°C/2hr   | 400.00 |                                  |                                  |
| MRC HEAT C      | 1500               | 10-12          | 2.10              | 110°C/2hr    | 350.00 | 50.0                             | 1.5                              |
|                 |                    |                |                   | 900°C/24hr   | 280.00 |                                  |                                  |
|                 |                    |                |                   | 130°C/2hr    | 480.00 |                                  |                                  |
| MRC HEAT K      | 1600               | 10-11          | 2.20              | 110°C/24hr   | 350.00 | 60.0                             | 1.0                              |
|                 |                    |                |                   | 900°C/2hr    | 290.00 |                                  |                                  |
|                 |                    |                |                   | 145°C/2hr    | 510.00 |                                  |                                  |
| MRC HEAT K(SPL) | 1600               | 10-11          | 2.10              | 110°C/24hr   | 450.00 | 62.0                             | 1.0                              |
|                 |                    |                |                   | 900°C/2hr    | 300.00 |                                  |                                  |
|                 |                    |                |                   | 1450°C/2hr   | 550.00 |                                  |                                  |
| MRC HEAT A      | 1750               | 9-10           | 2.8               | 110°C/24hr   | 600.00 | 86.0                             | 0.8                              |
|                 |                    |                |                   | 900°C/2hr    | 420.00 |                                  |                                  |
|                 |                    |                |                   | 1450°C/2hr   | 630.00 |                                  |                                  |

### Insulating Castables

| Product    | Max service Temp <sup>o</sup> C | Water addition | Dry Density gm/cc | CC5(kgf/cm <sup>2</sup> ) |         | Al <sub>2</sub> O <sub>3</sub> % | Fe <sub>2</sub> O <sub>3</sub> % |
|------------|---------------------------------|----------------|-------------------|---------------------------|---------|----------------------------------|----------------------------------|
| MRCRETE 11 | 1350                            | 35-40          | 1.25              | 110 <sup>o</sup> C/24 hr  | 700.00  | 45.0                             | 1.0                              |
|            |                                 |                |                   | 900 <sup>o</sup> C/2hr    | 800.00  |                                  |                                  |
|            |                                 |                |                   | 1100 <sup>o</sup> C/2hr   | 900.00  |                                  |                                  |
| MRCRETE 13 | 1350                            | 30-35          | 1.5               | 110 <sup>o</sup> C/24hr   | 750.00  | 60.0                             | 1.0                              |
|            |                                 |                |                   | 900 <sup>o</sup> C/2hr    | 810.00  |                                  |                                  |
|            |                                 |                |                   | 1100 <sup>o</sup> C/2hr   | 900.00  |                                  |                                  |
| MRCRETE 15 | 1350                            | 30-35          | 1.6               | 110 <sup>o</sup> C/2hr    | 800.00  | 80.0                             | 1.0                              |
|            |                                 |                |                   | 900 <sup>o</sup> C/24hr   | 900.00  |                                  |                                  |
|            |                                 |                |                   | 130 <sup>o</sup> C/2hr    | 1000.00 |                                  |                                  |

### Low Cement Castables

| Product | Max service Temp°C | Water addition | Dry Density gm/cc | CC5(kgf/cm²) |         | Al <sub>2</sub> O <sub>3</sub> % | Fe <sub>2</sub> O <sub>3</sub> % |
|---------|--------------------|----------------|-------------------|--------------|---------|----------------------------------|----------------------------------|
| LC 45   | 1550               | 5.0-6.0        | 2.3               | 110°C/24 hr  | 700.00  | 45.0                             | 1.0                              |
|         |                    |                |                   | 900°C/2hr    | 800.00  |                                  |                                  |
|         |                    |                |                   | 1100°C/2hr   | 900.00  |                                  |                                  |
| LC 60   | 1600               | 5.0-5.5        | 2.5               | 110°C/24hr   | 750.00  | 60.0                             | 1.0                              |
|         |                    |                |                   | 900°C/2hr    | 810.00  |                                  |                                  |
|         |                    |                |                   | 1100°C/2hr   | 900.00  |                                  |                                  |
| LC 80   | 1700               | 4.0-5.0        | 2.85              | 110°C/2hr    | 800.00  | 80.0                             | 1.0                              |
|         |                    |                |                   | 900°C/24hr   | 900.00  |                                  |                                  |
|         |                    |                |                   | 130°C/2hr    | 1000.00 |                                  |                                  |
| LC 90   | 1700               | 4.0-4.5        | 3.0               | 110°C/24hr   | 950.00  | 90.0                             | 1.0                              |
|         |                    |                |                   | 900°C/2hr    | 1000.00 |                                  |                                  |
|         |                    |                |                   | 1450°C/2hr   | 1100.00 |                                  |                                  |

## Gunning Mass :-

High Alumina Gunning Mass are used in Steel Ladles, EAF, LRF etc. Basic gunning Mass is used for EOF.

## Fire Clay & High Aluminam Mortar

| Product | Service Temp <sup>0</sup> C | Type of Setting | Al <sub>2</sub> O <sub>3</sub> % | Fe <sub>2</sub> O <sub>3</sub> % |
|---------|-----------------------------|-----------------|----------------------------------|----------------------------------|
| MR- 50  | 1550                        | Ceramic         | 50.0                             | 2.5                              |
| MR- 60  | 1600                        | Ceramic         | 60.0                             | 2.5                              |
| MR- 70  | 1650                        | Ceramic         | 70.0                             | 2.0                              |
| MR- 90  | 1700                        | Ceramic         | 90.0                             | 0.5                              |

## Basic Monolithics

| Product   | MgO % | SiO <sub>2</sub> % | Grading mm | Sintering (Temp <sup>0</sup> C) | Application (Temp <sup>0</sup> C) |
|-----------|-------|--------------------|------------|---------------------------------|-----------------------------------|
| MR GUN 80 | 80.00 | 7.0                | 0-3        | 1500                            | 1700                              |

## Blast Furnace Anhydrous Tap Hole Mass / Anhydrous Mudgun Mass :-

It is specialised product for different capacities blast furnaces

|                             |                                                                                      |                                                                                                       |
|-----------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Tap Hole Mass (Resin)       | Al <sub>2</sub> O <sub>3</sub> =46%<br>Sic+C=28%<br>Resin=6%<br>Grain Size=0.3 mm    | B.D at 110 <sup>0</sup> C/24 hrs=2.2 gm/cc<br>CCS at 900 <sup>0</sup> C/2 hrs=60 kg f/cm <sup>2</sup> |
| Tap Hole Mass (Tar & Resin) | Al <sub>2</sub> O <sub>3</sub> =40-44%<br>Sic+C=26%<br>Resin=2%<br>Grain Size=0-3 mm | B.D at 110 <sup>0</sup> C/24 hrs=2.0 gm/cc<br>CCS at 900 <sup>0</sup> C/2 hrs=60 kg f/cm <sup>2</sup> |



## OUR PRODUCT

### Basic Monolithics :-

| Product   | MgO%  | SiO <sub>2</sub> % | Grading mm | Sintering (Temp°C) | Application (Temp°C) |
|-----------|-------|--------------------|------------|--------------------|----------------------|
| MR RAM-85 | 85.00 | 6.0                | 0-5        | 1550               | 1700                 |
| MR RAM-86 | 86.00 | 6.0                | 0-5        | 1550               | 1700                 |
| MR RAM-80 | 80.00 | 7.0                | 0-3        | 1500               | 1700                 |

### Ramming Masses :-

Basic ramming mass is used for EAF heat cold and hot steel launders. Alumina ramming masses are available for induction, burner block etc. Alumina based tar and resin bonded ramming masses are available for Blast Furnace Through. Blast Furnace Tap Hole Mass:- It is a special product for di Hereneapeities blast furnaces. Basic Ramming Masses :-

| Product   | MgO%  | SiO <sub>2</sub> % | Grading mm | Sintering (Temp°C) | Application (Temp°C) |
|-----------|-------|--------------------|------------|--------------------|----------------------|
| MR RAM-85 | 85.00 | 6.0                | 0-5        | 1550               | 1700                 |
| MR RAM-86 | 86.00 | 6.0                | 0-5        | 1550               | 1700                 |

### Blast Furnace Anhydrous Runner Mass/Trough Mass :-

Both Resis & Tar Bonded and special type Resin bonded Runner Mass/Trough Mass are available for all capacities blast furnaces in India. MRAI is having the potential customers for supplying Tap hole Clay for the Blast Furnace of Capacity of 40 m3 - 4000 m3 (Volume) all over India.

|                  |                                |            |                                                |
|------------------|--------------------------------|------------|------------------------------------------------|
| Anhydrous        | Al <sub>2</sub> O <sub>3</sub> | = 58-62%   | B.D. at 110° C/24 hrs = 2.2-2.4 gm/cc          |
| Trough Mass      | Sic+C                          | = 20-24%   | CCS at 110° C/24 hrs = 160 kgf/cm <sup>2</sup> |
| Resin+Tar Bond   | Grain Size                     | = 0-5 mm   |                                                |
| Anhydrous        | Al <sub>2</sub> O <sub>3</sub> | = 62-66%   | B.D. at 110° C/24 hrs = 2.4-2.8 gm/cc          |
| Trough Mass      | Sic                            | = 20%(Min) | CCS at 110° C/24 hrs = 200 kgf/cm <sup>2</sup> |
| Spcl. Type Resin | Grain Size                     | = 0-5 mm   |                                                |

## Research & Development

Continuous and incremental development in the R&D is main objective to improve the existing products and process besides new products development.

MRAI had invested a well-equipped lab instruments for incoming raw material Quality checking like :

- a) Chemical Testing facilities for incoming Raw materials.
- b) Distillation Apparatus for coal Tar testing.
- c) Ring & Ball apparatus.
- d) Porosity Measurement Apparatus.
- e) Mould Vibrator for castable.
- f) High capacity Muffle Furnace.
- g) Sand Rammer for Workability Testing.
- h) C.C.S Machine.



MRAI is having huge storage area of finished good at Plant Precises so that proper aging period is computed before dispatch of the product . This will increase the acceptibility of product by customers



## MRAI

### Quality:

We should always be Quality Conscious for Betterment of our Products. We always take care refractory products material that can withstand high temperature, mechanical wear & chemical attack. They are critical for steel industries also ability to perform reliably under specific service condition.

### Customer Satisfaction :

MRAI is always committed to satisfy Customers by supplying Quality Products by understanding customer need and obtaining timely feedback.

### Our Goal :

Our goal is to be the leading Manufacturing Unit of Refractory Products in Domestic & International Market. The Goal is to deliver reliable, high – performance refractory solutions that meet or exceed customer expectations in service life, performance and value ensuring maximum customer satisfaction and long-term trust.

### Services :

Our motto is total customer satisfactions using our technical services as it does from our product quality. We have a good and healthy marketing and engineering support to provide value added services and ensure technically correct and cost effective installation of the Refractory Monolithic.

MRAI used to take consultation from expert Chattered Accountants , Technocrats / Technical Advisors, Experienced Ceramic experts & Legal advisors time to time for strengthening plant operation & management.

## OUR PRODUCT

### Other Services :-

Misus Refractories and Allied Industries also provides through mass application and maintenance services for Blast Furnace Runners.

Our specialized team of expert technicians have huge experience in design of runner & application of runner mass in different plants.

### Our Services include :-

1. Complete Design of blast furnace runner as per furnace parameters.
2. Supply of Runner Mass for application.
3. Application of Runner mass for Main Runner, Mental Runner & Slag Runner
4. Services such as maintenance, Application & monitoring of performance of runner (24 x 7) for main Runner, Metal Runner, Metal Runner & Slag Runner.

### Actual Work Image :-



## MRAI MARKETING NETWORK

**Our Esteem Clients are in following States :**

- 1) West Bengal**
- 2) Jharkhand**
- 3) Chhattisgarh**
- 4) Odisha**
- 5) Goa**
- 6) Andhra Pradesh**
- 7) Karnataka**
- 8) Maharashtra**
- 9) Gujarat**



## COMMUNITY DEVELOPMENT PROGRAMS

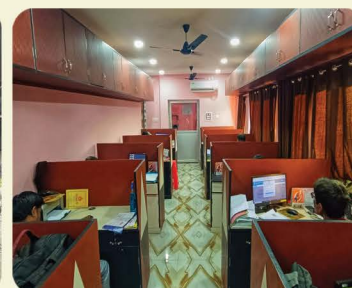
Since inception, Community Development programs has been intrinsic part of programs strategy, It includes number of initiatives focused on rural areas like development & support to schools. Classrooms and Library was developed at Tajpur high school for development of students.

MRAI also contributes & takes participation in local social functions like Puja Celebrations and Blood donation camp etc. MRAI had also organised Vaccination Camp (During Covid-19 Period) for welfare of workers and local community.

A blood donation Camp was organised at Factory premises through Barjora Super Speciality Hospital. The employes were highly enthusiastic to donate the blood for noble cause.



# PHOTO GALLERY



## PHOTO GALLERY



# TESTIMONIAL



## OUR CLIENTS



सेल SAIL



TATA STEEL



Evonith



ELECTROSTEEL



WELSPUN INDIA



JINDAL



ArcelorMittal



NIPPON STEEL



JINDAL SAW LTD.  
TOTAL PIPE SOLUTIONS



Jai Balaji Group



KALYANI STEEL



KALYANI



Rungta Mines Limited  
Chaibasa

RUNGTA STEEL®

## CONTACT US

### Work & Office :

**MISUS REFRACTORIES & ALLIED INDUSTRIES**

**Hat Ashuria More, Barjora,**

**P.O. U-jambedia**

**Dist: Bankura, West Bengal,**

**INDIA**

**PIN-722202**

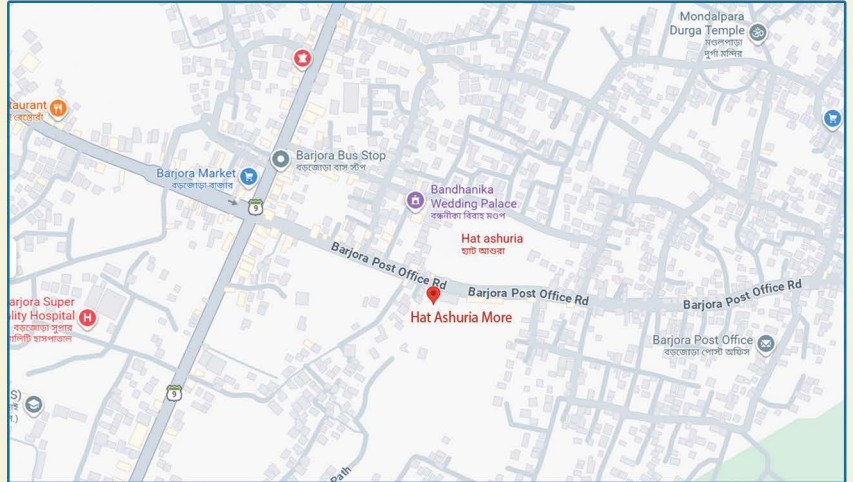
**[misusbarjora@yahoo.com](mailto:misusbarjora@yahoo.com)**

**Mr. Monojit Sengupta**

**Chief Executive Officer**

**Ph: 09434216176, 09332297691**

**[misusbarjora@yahoo.com](mailto:misusbarjora@yahoo.com)**



### Communication for Trade :

**Mr. Sourav Sarkar**

**Business Consultant**

**Ph : 08879295876**

**[souravivasarkar@rediffmail.com](mailto:souravivasarkar@rediffmail.com)**

**Mr. Prasanta Thakura**

**Dy General Manager (Marketing & Application)**

**Ph : 09332340870, 09547046762**

**[pthakuradisha@gmail.com](mailto:pthakuradisha@gmail.com)**

**Any Type of Communication**

**Mr. Sukumar Biswas**

**Head - HR & Administration**

**Ph: 08798229575**

**[misusbarjora@yahoo.com](mailto:misusbarjora@yahoo.com)**





**MISUS REFRACTORIES & ALLIED INDUSTRIES**  
( ISO 9001:2015 & ISO 45001:2018 CERTIFIED )  
Hat Ashuria More , Barjora , Bankura , Pin. 722202

**[misusbarjora@yahoo.com](mailto:misusbarjora@yahoo.com)**